Journal of Linguistics (2026), **0**: 1–12
doi:10.1017/S0022226721000XXX

Journal of Linguistics

CAMBRIDGE UNIVERSITY PRESS

RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

Author[1] and Author[1]

[1]University, Address, City, Country.

Received: 01 Jan 2026; **Revised:** 02 Jan 2026; **Accepted:** 03 Jan 2026

Keywords: Hungarian, syntax, focus, exhaustivity, contrastivity, visual-world paradigm, sentence–picture verification, self-paced reading, eye-tracking

Abstract

Abstract goes here. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Contents

1 Introduction **2**
2 Methods **5**
2.1 Participants . . . 5
2.2 Materials . . . 6
2.3 Design . . . 7
2.4 Procedure . . . 8
2.5 Data Analysis . . . 8
2.6 Analysis . . . 9
3 Results **9**
4 Discussion **10**
5 Conclusion **10**

1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eyetracking and the visual-world paradigm. Specifically, we examine two theoretically important inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain exhaustively, and whether contrastive but non-exhaustive readings exist, as well as from a cognitive perspective, whether different focus types produce processing differences during language comprehension.

(1) *Anna* *meg-ette* *a* *banánt.*
Anna[NOM] VPX-eat-PST-3SG.DEF the banana-ACC
'Anna ate (up) the banana.'

(2) *Anna* *a* *BANÁNT* *ette* *meg.*
Anna[NOM] the banana-ACC eat-PST-3SG.DEF VPX
a) 'Anna ate (up) the BANANA.' [and nothing else]
b) 'Anna ate (up) the BANANA.' [and not something else]

Consider the canonical, neutral sentence in example 1, where the subject *Anna* is in the sentence-initial position, followed by the verb, *ette* 'ate' (together with the verbal prefix [VPX] (or verbal modifier [VM]) *meg-*)[1] and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example 2, where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind. The sentence in example 2 is a typical example for a Hungarian focus construction, and it could also be rendered in English with an *it*-cleft, i.e., 'It is the BANANA[2] that Anna ate.'. The comparison of the Hungarian focus-constructions with English *it*-clefts has its own literature (cf. É. Kiss 1999; Wedgwood et al. 2006), but it is out of the scope of the present discussion.

Constituents in canonical sentences can be focused for various communicative purposes, such as emphasis, question answering, or contradiction. In 2 we are focusing the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position (Balogh 2007; Wedgwood 2006) or structural focus position (É.Kiss 1998) or simply just focus position – often associated with exhaustive/identificational semantics (Balogh 2013). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary (Balogh 2013). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. Gécseg 2020; Kornai & Kálmán 1988), meaning that the verb is destressed and the acoustic prominence of all other elements are leveled. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. É. Kiss 1995) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research.

[1] Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating that the action has been completed.
[2] Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

One of the key issues around Hungarian focus is whether it is inherently exhaustive, or if the exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in 2 above, does this sentence implies that Anna ate only the banana *and nothing else* – an exclusive reading, or could it mean that Anna ate the banana *and not something else* (or even, that she ate the banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the 'default' interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively without additional context, as demonstrated by the two possible readings given in 2.

We should therefore try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal structure, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)'s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we will argue that identificational focus does not always entail exhaustivity. According to (Rooth 1992)'s theory on focus interpretation, exhaustivity is not inherent to focus but arises when focus interacts with specific operators like 'only'. For him, focus is a mechanism that introduces a set of alternatives (*alternative semantics*), and he considers contrast to be the pragmatic use of focus, where one alternative is highlighted against others in the set. In Vallduví & Vilkuna (1998), *kontrast* is an operator-like category that introduces a set of alternatives for the focused constituent. If an expression a is contrastive, a membership set $M = \{\ldots, a, \ldots\}$ is generated, serving as a quantificational domain. This allows alternative propositions $P(x)$ to be derived by substituting other members of M for a. Thus, *kontrast* provides the semantic mechanism by which focus evokes alternatives, enabling identificational or exhaustive readings. In Krifka (2008:259)'s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka's distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus. Zimmermann (2008) also argued that contrastivity is not a semantic property of focus at all but rather a pragmatic one, and that it is best understood in terms of hearer expectation and discourse predictability, as a discourse-semantic device that signals unexpectedness relative to the hearer's assumptions, rather than a purely semantic operator over alternatives.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikeliness, or contradicting nature given the context, e.g., 'Anna ate the banana [and not the apple].', but it does not necessarily exclude alternatives, i.e., there could be other things that Anna ate, e.g., 'Anna ate the banana [and/but also the apple].' Now consider the following:

(3) *Anna* ***csak*** *a* *BANÁNT* *ette* *meg.*
Anna[NOM] only the banana-ACC eat-PST-3SG.DEF VPX
'Anna ate (up) only the BANANA.'

(4) *Anna* ***többek között*** *a* *BANÁNT* *ette* *meg.*
Anna[NOM] others amongst the banana-ACC eat-PST-3SG.DEF VPX
'Anna ate (up) the BANANA, amongst other things.'

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* 'only' to the sentence, as in example 3[3]. One intuitive reading of a simple PVF sentence like that in example 2 would be that 'Anna ate the banana [and nothing else].' In this sense, the sentence seems to be equal to that in 3, where the exhaustive interpretation is explicitly expressed with the operator *csak* 'only'. If we accept É.Kiss (1998)'s view, this seems to be redundant as the structural focus is already exhaustive. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007).

Moreover, we can also cancel exhaustivity within the same clause with the help of certain phrases, such as *többek között* 'amongst others', as in example 4[4], by inserting a phrase with an inherently non-exhaustive/inclusive meaning (see also Balogh 2013; Wedgwood 2006:14). This sentence is not only perfectly grammatical, inclusive, but also expresses a contrastive sense due the use of focus (there must be some specific discourse function for choosing to focus a constituent).

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981) (the received view up until the 2000s). More recently, this view has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that 'an immediately pre-verbal focus in Hungarian is semantically exhaustive' or 'is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like "only"', in their first example[5] (analogous to our example 2) Onea & Beaver (2009:1) suggested that the implicit reading of a sentence of this type of construction should be exhaustive, i.e. 'Anna ate the banana [and nothing else].' but their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

Looking at example 2 however, we feel that the underlying implication is not necessarily that Anna ate *nothing* else, but that she did not eat *something* else. Therefore, in our view, the default reading might as well be contrastive.

[3] The scope of *csak* 'only' only extends to the subsequent element, it can only refer to the object and not the following VP.

[4] Similarly, only the object is in the scope of *többek között* 'amongst others', it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

[5] *Péter MARIT csókolta meg.* Peter Mari-ACC kiss-PST-3SG.DEF VPX 'Peter kissed MARY.'

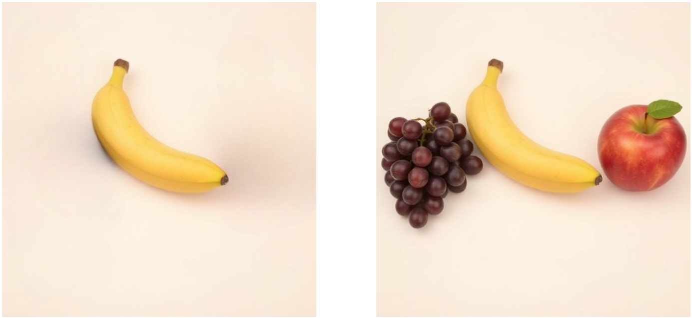

Figure 1. *Example stimuli for a sentence–picture verification task*

To test this, we designed an experiment within the visual world paradigm where people are first presented with typical PVF sentences, and then given a binary choice that offers two contexts where both are correct and plausible: either A) a single object image corresponding to an exhaustive interpretation, or B) a multi-object image implying a contrastive interpretation (see the image pair in Figure 1 accompanying examples 1-4). Participants have to choose the image that best matches the sentence they just read in a sentence–picture verification task. Their choices should tell us about the preferred interpretation of the PVF constructions, and whether it is more naturally interpreted as exhaustive or contrastive. Besides collecting behavioural data on the choices participants made, we also recorded their reading times and collected eye-tracking data during the picture verification phase, which hopefully can give us additional insights into the cognitive processes involved in interpreting the different types of PVF constructions.

To sum up, while previous work has mainly tried to answer to whether an immediately pre-verbal focus in Hungarian is semantically exhaustive or not, our question is: What is the default interpretation of a pre-verbal focus construction? Is it inherently exhaustive, or is it more naturally contrastive?

2. Methods

To address the theoretical debate, and in light with recent experimental studies (Gerőcs et al. 2014; Káldi et al. 2021; Onea & Beaver 2009), we designed a visual-world experiment with eye-tracking to differentiate interpretations associated with the PVF structures. Specifically, we were curious about Hungarian speakers' dominant interpretation of the unmodified, simple PVF sentence (e.g., 2), in comparison with the *only*–exhaustive (3) and the *amongst others*–contrastive conditions (4).

2.1. Participants

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data), the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (unusually long submission times were rejected).

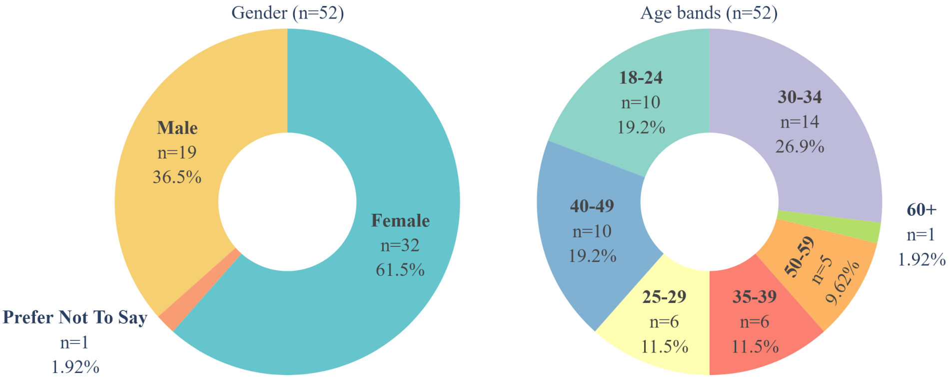

Figure 2. *Demographic information of the participants in the experiment; gender (left) and age distributions (right).*

2.2. Materials

Our stimuli consisted of 54 experimental sentences (18 items × 3 conditions) with 54 additional filler sentences in a randomized order, but making sure that the same items never followed each other. Regarding lexical variation, we used verbs both with and without verbal prefixes (50-50%), and we included instances of both definite and indefinite conjugation (44.4-55.6%); also, both countable and uncountable nouns were used for object focus. We added some simple contextual fillers to the beginning of each sentence, such as *ami X-t illeti* 'regarding X' or time or place adverbials to balance sentence length and make the targeted focus constructions appear not in the beginning of sentences/trials. When presenting the sentences on the screen, we divided them into regions, and articles were kept together with nouns when presented on the screen, so all three conditions had the same number of regions (7), and the focused element was always in region 6 (r6). The stimuli were created by modifying a set of 18 base sentences, which were all simple transitive sentences with a subject, verb, and direct object. The three conditions were created by modifying the base sentences with the addition of the focus marker *csak* 'only' for the exhaustive condition, and *többek között* 'amongst others' for the contrastive condition...

The 18 items were accompanied by 18 image pairs, these were created using DeepAI's image generator (https://deepai.org/) with prompts requesting simple 3D objects & scenarios in front of a light pastel background, and then modified manually to get matching pairs. The full stimuli list (sentences + visuals) is available online ??.

To avoid overly long experimental sessions, we divided the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap. This resulted in 72 valid submissions (25 for group one, 23 for group two, and 24 for group three), thus every stimuli was interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

The three conditions were: a) exhaustive focus with *csak* ‘only’ (e.g., 5a), b) unmodified, simple pre-verbal focus (PVF) (e.g., 5b), and c) contrastive focus with *többek között* ‘amongst others’ (e.g., 5c). Each item was paired with two images: A) one depicting a single-object interpretation – exhaustive reading – and B) one depicting a multiple-object interpretation – contrastive reading. See Figure 3 for an example of the image pairs accompanying the sentences in the experiment.

(5) a. **exhaustive focus**

Ami	*az*	*innivalót*	*illeti,*	*Eszter*	*csak*	*kávét*	*ivott.*
which[REL]	the	drink-ACC	regard-3SG	Eszter	only	coffee-ACC	drink-PST-3SG

‘Regarding drinks, Eszter only drank coffee.’

b. **unmodified pre-verbal focus**

Ami	*az*	*innivalót*	*illeti,*	*Eszter*	*ma*	*kávét*	*ivott.*
which[REL]	the	drink-ACC	regard-3SG	Eszter	today	coffee-ACC	drink-PST-3SG

‘Regarding drinks, Eszter drank coffee today.’

c. **contrastive focus**

Ma	*reggel*	*Eszter*	*többek*	*között*	*kávét*	*ivott.*
today	morning	Eszter	others	amongst	coffee-ACC	drink-PST-3SG

‘This morning, Eszter drank, amongst other things, coffee.’

Figure 3. *Image pair for the sentence in examples 5a, 5b, & 5c*

2.3. Design

We used a sentence–picture verification (SPV) task, in which participants read a sentence in a self-paced reading (SPR) manner and were asked to indicate one correct meaning-representation between two visual stimuli.

After a practice session consisting of three trials, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen’s width and 40% of the screen’s height) in a canvas 50% larger than the images.

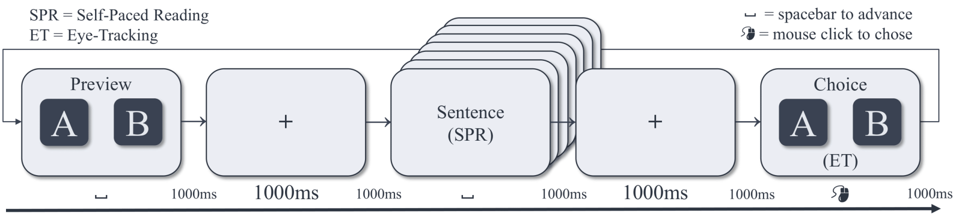

Figure 4. *Design*

After pressing the spacebar they were presented with the sentence word by word (region by region), in the middle of the screen in a self-paced reading (SPR) manner participants had to use the spacebar to advance throught the sentences. We recorded reaction times (RT) throughout the reading, and the focused elements appeared in region six (r6) in every condition. After reading through the sentence, the image pair re-appeared and participants were asked to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas. Reaction times and eye movements were recorded during this picture verification phase.

Eye tracking what?

Images type A and B were counterbalanced throughout the trials.

2.4. Procedure

Participants completed the experiment on their own devices (PC only with a webcamera and physical keyboard & mouse) and own their own time, but with Prolific helping us control for too quick or too slow submissions resulting in rejection/timeout.

Participants were instructed to be in a well-lit environment, sit at a comfortable distance from their screen and webcam, and minimize head movements during the experiment, and were informed about the exact procedure.

Participants also had to calibrate the eye-tracker before starting the experiment After calibration, practice trials were provided to familiarize participants with the self-paced reading and picture verification tasks.

Eye-tracking data were collected using the WebGazer.js library built in PCIbex, which uses the participant's webcam to track eye movements. Before starting the experiment, participants underwent a calibration procedure to ensure accurate eye-tracking data collection.

2.5. Data Analysis

Reading times and dwell times were analyzed using linear mixed-effects models with FocusCondition, ChoiceType, and their interaction as fixed effects, and Participant and Item as random effects. Post-hoc comparisons were conducted using Tukey HSD tests. Trials with incorrect responses or extreme RTs (>2.5 SD from the mean) were excluded.......

2.6. Analysis

3. Results

The behavioural results showed that participants overwhelmingly interpreted the unmodified PVF sentences like only-sentences, choosing type A images, but in the "contrastive" condition (among othersentences) they preferred the multi-object type B images, exhibiting a tendency toward non-exhaustive readings (Table 1). To further examine how the focus structures were associated with exhaustive and contrastive readings, we fitted a linear mixed-effects model to the total dwell time on images, with FocusCondition, ChoiceType and their interaction as main effects, and participant and item as random effects. The results revealed significant effects of FocusCondition ($\chi^2(2) = 11.67$, $p < .001$), ChoiceType ($\chi^2(1) = 19.51$, $p < .001$), and interaction between FocusCondition and ChoiceType ($p < .001$). Interestingly, post-hoc (TukeyHSD) tests showed that after reading only-sentences, participants took significantly less time to choose A compared to B ($\beta = –0.88$, $p < .001$), whereas they spent significantly more time to select A after reading contrastive sentences ($\beta = .21$, $p = .034$), suggesting that participants were faster to pick the expected choices in the "exhaustive" and "contrastive" conditions, respectively. Crucially, no significant difference was observed between selecting A or B for the simple PVF sentences ($\beta = -.19$, $p = .362$), implying that participants might consider both representations equally plausible when making their choices. Overall, our results suggest that Hungarian pre-verbal focus constructions tend to be interpreted exhaustively when no additional context is provided. This exhaustive bias seems to persist even when 'among others' is present, despite its licensing a contrastive reading in truth value. This explains the apparent tension in theoretical accounts. Our results also showed facilitation for explicitly exhaustive sentences and not for simple PVFs, consistent with online disambiguation of an underspecified structure.

Table 1. *Type of chosen image per condition.*

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	93.9 (169)	6.1 (11)
PVF	94.4 (170)	5.6 (10)
Contrastive	41.7 (75)	58.3 (105)

Table 2. *Total dwell time during choice (ms).*

Condition	Image A Mean (SD)	Image B Mean (SD)
Exhaustive	1054 (1154)	2173 (2001)
PVF	1015 (991)	1844 (1939)
Contrastive	1206 (1323)	1226 (1157)

Table 3. *Type of chosen image per condition.*

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	93.9 (169)	6.1 (11)
PVF	94.4 (170)	5.6 (10)
Contrastive	41.7 (75)	58.3 (105)

Table 4. *Total dwell time during choice (ms).*

Condition	Image A Mean (SD)	Image B Mea
Exhaustive	1054 (1154)	2173
PVF	1015 (991)	1844
Contrastive	1206 (1323)	1226

Table 5. *Chosen image per condition.*

Cond.	A % (N)	B % (N)
Exhaustive	93.9 (169)	6.1 (11)
PVF	94.4 (170)	5.6 (10)
Contrastive	41.7 (75)	58.3 (105)

Table 6. *Dwell time during choice (ms).*

Cond.	A Mean (SD)	B Mean (SD)
Exhaustive	1054 (1154)	2173 (2001)
PVF	1015 (991)	1844 (1939)
Contrastive	1206 (1323)	1226 (1157)

4. Discussion

5. Conclusion

…

Unnumbered section

Acknowledgments. The authors also wish to thank Lucie Laporte-Devylder, Jordan Zlatev, Georgios Stampoulidis, Joost Van de Weijer, Katie Hoemann, and two anonymous reviewers for their invaluable feedback and comments on earlier versions of this paper.

Funding statement. This study was funded by the Deutsche Forschungsgemeinschaft (DFG) through the Collaborative Research Center 1252 Prominence in Language, and by Mobility Grants from Division 7, Research Management, University of Cologne, which we gratefully acknowledge.

Data availability statement. A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings – e.g. Replication data and code can be found in Harvard Dataverse: https://doi.org/link.

References

Balogh, Kata. 2007. Focus and 'Only' in Hungarian. In Balder D. Ten Cate and Henk W. Zeevat (eds.), *Logic, Language, and Computation* 4363, 31–44. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-540-75144-1_3 (accessed 2026 February 2).

Balogh, Kata. 2013. Hungarian Pre-verbal Focus and Exhaustivity. In Yoichi Motomura, Alastair Butler, and Daisuke Bekki (eds.), *New Frontiers in Artificial Intelligence: JSAI-isAI 2012 Workshops, LENLS, JURISIN, MiMI, Miyazaki, Japan, November 30 and December 1, 2012, Revised Selected Papers* (Lecture Notes in Computer Science 7856), 1–16. Berlin; Heidelberg: Springer. https://doi.org/10.1007/978-3-642-39931-2 (accessed 2026 February 2).

É. Kiss, Katalin (ed.). 1995. *Discourse Configurational Languages* (Oxford Studies in Comparative Syntax 4). New York: Oxford University Press.

É. Kiss, Katalin. 1998. Identificational Focus versus Information Focus. *Language* 74(2), 245–273. https://www.jstor.org/stable/417867 (accessed 2026 February 2).

É. Kiss, Katalin. 1998. Multiple Topic, One Focus? *Acta Linguistica Hungarica* 45(1/2), 3–29. https://www.jstor.org/stable/44656166 (accessed 2026 February 2).

É. Kiss, Katalin. 1999. The English Cleft Construction as a Focus Phrase. In Lunella Mereu (ed.), *Boundaries of Morphology and Syntax* (Current Issues in Linguistic Theory 180), 217. Amsterdam; Philadelphia: John Benjamins Publishing Company. https://doi.org/10.1075/cilt.180.14kis (accessed 2026 February 2).

É. Kiss, Katalin. 2007. Topic and Focus: Two Structural Positions Associated with Logical Functions in the Left Periphery of the Hungarian Sentence. *Acta Linguistica Hungarica* 55(3–4), 287–296. https://doi.org/10.1556/aling.55.2008.3-4.4 (accessed 2026 February 2).

Gécseg, Zsuzsanna. 2020. Focused Topics in Hungarian: Reconsidering the Pragmatic Relevance of the Structural Focus Position. *Studia Linguistica* 74(3), 740–774. https://doi.org/10.1111/stul.12142 (accessed 2026 February 2).

Genzel, Susanne, Shinichiro Ishihara, and Balázs Surányi. 2015. The prosodic expression of focus, contrast and givenness: A production study of Hungarian. *Lingua* 165, 183–204. https://doi.org/10.1016/j.lingua.2014.07.010 (accessed 2026 February 2).

Gerőcs, Mátyás, Anna Babarczy, and Balázs Surányi. 2014. Exhaustivity in focus: Experimental evidence from Hungarian. In Joseph Emonds and Markéta Janebová (eds.), *Language Use and Linguistic Structure Proceedings of the Olomouc Linguistics Colloquium 2013*, 181–195. Olomouc, Czech Republic: Palacký University. https://core.ac.uk/reader/42931210 (accessed 2026 February 2).

Horvath, Julia. 2008. Separating "Focus Movement" from Focus. In Simin Karimi, Vida Samiian, and Wendy K. Wilkins (eds.), *Phrasal and Clausal Architecture: Syntactic Derivation and Interpretation. In Honor of Joseph E. Emonds*, 108–145. Amsterdam; Philadelphia: John Benjamins Publishing Company. https://www.degruyterbrill.com/document/doi/10.1075/la.101.07hor/html (accessed 2026 February 7).

Hsu, Yu-Yin. 2019. Associations between Focus Constructions and Levels of Exhaustivity: An Experimental Investigation of Chinese. *PLOS ONE* 14(10), e0223502. https://doi.org/10.1371/journal.pone.0223502 (accessed 2026 February 2).

Hsu, Yu-Yin. 2024. Modal Raising and Focus Marking in Chinese. *Journal of Linguistics* 60(3), 635–666. https://doi.org/10.1017/S0022226723000142 (accessed 2026 February 2).

Káldi, Tamás and Anna Babarczy. 2016. A magyar fókusz és a skaláris implikatúrák : egy szemmozgás-követéses kutatás eredményei [Hungarian focus and scalar implicatures: Results of an eye-tracking study]. In Bence Kas (ed.), *Szavad ne feledd!* 333–346. Budapest: MTA Nyelvtudományi Intézet. https://real.mtak.hu/48340/ (accessed 2025 November 21).

Káldi, Tamás. 2021. H*ungarian Pre-verbal Focus: Representation and Interpretation.* Ph.D. dissertation, Budapest University of Technology and Economics. https://www.researchgate.net/publication/353145107 (accessed 2026 February 2).

Káldi, Tamás; Szöllösi, Ágnes; Babarczy, Anna. 2021. Hungarian Structural Focus: Accessibility to Focused Elements and Their Alternatives in Working Memory and Delayed Recognition Memory. *Frontiers in Psychology* 12, 514886. https://doi.org/10.3389/fpsyg.2021.514886 (accessed 2026 February 2).

Kenesei, István. 2006. Focus as Identification. In Valéria Molnár and Susanne Winkler (eds.), *The Architecture of Focus*, 137–168. Berlin: Walter de Gruyter. (accessed 2026 February 2).

Kornai, András; Kálmán, László. 1988. Hungarian sentence intonation. In Harry van der Hulst and Norval Smith (eds.), *Autosegmental studies on pitch accent*, 183–195. Dordrecht: Foris.

Krifka, Manfred. 2008. Basic Notions of Information Structure. *Acta Linguistica Hungarica* 55(3–4), 243–276. https://doi.org/10.1556/ALing.55.2008.3-4.2 (accessed 2026 February 2).

Onea, Edgar. 2009. Exhaustiveness of Hungarian Focus: Experimental Evidence from Hungarian and German. In Arndt Riester and Edgar Onea (eds.), *Focus at the Syntax-Semantics Interface* (Working Papers of the SFB 732 3), 53–68. University of Stuttgart (accessed 2026 February 2).

Onea, Edgar; Beaver, David. 2009. Hungarian Focus Is Not Exhausted. In *Proceedings of Semantics and Linguistic Theory (SALT 19)*, 342–359. https://doi.org/10.3765/salt.v19i0.2524 (accessed 2026 February 2).

Rooth, Mats. 1985. *Association with Focus.* Ph.D. dissertation, University of Massachusetts. https://ecommons.cornell.edu/entities/publication/b16d4b65-30ea-4b87-9857-788aa69dc393 (accessed 2026 February 2).

Rooth, Mats. 1992. A Theory of Focus Interpretation. *Natural Language Semantics* 1(1), 75–116. https://doi.org/10.1007/BF02342617 (accessed 2026 February 2).

Schwarz, Florian; Zehr, Jonathan. 2021. Tutorial: Introduction to PCIbex – An Open-Science Platform for Online Experiments: Design, Data-Collection and Code-Sharing. *Proceedings of the Annual Meeting of the Cognitive*

Science Society 43, 15–16. https://escholarship.org/uc/item/1ng1q4c6 (accessed 2026 February 2).

Szabolcsi, Anna. 1981. Compositionality in Focus. *Folia Linguistica* 15(1–2), 141–161. https://doi.org/10.1515/flin.1981.15.1-2.141 (accessed 2026 February 2).

Vallduví, Enric; Vilkuna, Maria. 1998. On Rheme and Kontrast. In Peter Culicover and Louise McNally (eds.), *The Limits of Syntax*, 79–108. Leiden: Brill. https://doi.org/10.1163/9789004373167_005 (accessed 2026 February 2).

Wedgwood, Daniel. 2006. Predication, Focus and the Positions of Negation in Hungarian. *Lingua* 116(3), 351–376. https://doi.org/10.1016/j.lingua.2004.08.009 (accessed 2026 February 2).

Wedgwood, Daniel; Petho, Gergely; Cann, Ronnie. 2006. Hungarian 'Focus Position' and English It-Clefts: The Semantic Underspecification of 'Focus' Readings. https://www.lel.ed.ac.uk/~ronnie/JoS-paper_forESRC.pdf (accessed 2026 February 2).

Zimmermann, Malte. 2008. Contrastive Focus and Emphasis. *Acta Linguistica Hungarica* 55(3–4), 347–360. https://doi.org/10.1556/ALing.55.2008.3-4.9 (accessed 2026 February 2).